Model Predictive Control
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Recitation 04
Optimization

Exercise 1

Are the following sets convex? Justify your answers.

$P_1 = \{x \in \mathbb{R}^n \mid Ax \leq b\}$
$P_2 = \{x \in \mathbb{R}^n \mid x^\top Hx \leq 2, H \succ 0\}$
$P_3 = \{x \in \mathbb{R}^n \mid x^\top x = 0\}$
$P_4 = \{x \in \mathbb{R}^n \mid x^\top x = 2\}$
Set of non-negative functions: $P_5 = \{f(x) | f(x) \geq 0 \ \forall x \in \mathbb{R}^n\}$
Set of positive semi-definite matrices: $P_6 = \{M \in \mathbb{R}^{n\times n} | M^\top = M, \ x^\top Mx \geq 0 \ \forall x \in \mathbb{R}^n\}$

Exercise 2

Consider the optimization problem

$$
\begin{aligned}
P: \quad x_p^\star = \arg \min_{x\in\mathbb{R}^2} \ & \|x\|_p \\
\text{s.t. } & x_1 + x_2 = 3 \\
& (x_2 - 2)^2 \leq 1
\end{aligned}
$$

with $x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$.

i) Draw the feasible set of P.

ii) State the set of all optimal solutions $X_1^\star$ for $p = 1$, $X_2^\star$ for $p = 2$, and $X_\infty^\star$ for $p = \infty$.

Exercise 3

Show that the duality gap is zero and the KKT conditions are satisfied for the following convex optimization problem

$$
\begin{aligned}
\min_{x\in\mathbb{R}^2} \quad & c^T x \\
\text{subj. to} \quad & x_1 \geq 1 \\
& x_2 \geq 2
\end{aligned}
$$

with $c = [3, 4]^T$.

Exercise 4

Is $f(x) = \min\{3x, 2x\}$ a convex function? Justify your answer.